# CCTV Gun Detection - Sprint 3 Summary

Project: CCTV Gun Detection - Reproduction Study

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## 1. Sprint 3 Goal

The goal of Sprint 3 was to move from pipeline preparation to baseline training and initial results. The main objectives were to train full local runs of YOLO11n and YOLO26n under the official project split, compare their validation behavior, inspect error patterns with confusion matrices, and prepare the first test-set evaluation artifacts.

## 2. Experimental Protocol and Team Contributions

The Sprint 2 protocol was kept fixed in order to preserve methodological consistency: Cam1 and Cam7 were used for train/validation, Cam5 was kept as test, handgun and short_rifle were mapped to a single weapon class, and knife was excluded from the initial detection setup. Matheus led baseline training, local run execution, and result collection. Fernanda focused on comparison analysis, model interpretation, and checkpoint material preparation. The main experimental decisions and result interpretation were shared by the team.

- Official split preserved from Sprint 2
- Same YOLO-formatted dataset used for both models
- Local full training runs for YOLO11n and YOLO26n
- Validation and test confusion matrices collected from local artifacts

## 3. Full Local Training Results

Two full local runs were completed: YOLO11n and YOLO26n. For each model, the main validation checkpoint was selected using the best mAP50-95 value. The resulting metrics are shown below.

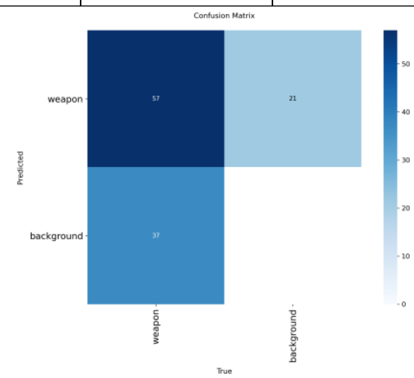
| Model | Best epoch | Precision | Recall | mAP@50 | mAP@50:95 |
|---|---|---|---|---|---|
| YOLO11n | 82 | 72.7% | 59.6% | 63.0% | 32.5% |
| YOLO26n | 71 | 82.8% | 60.6% | 68.5% | 36.7% |

Under this local protocol, YOLO26n outperformed YOLO11n on the validation split. It achieved higher mAP@50 (68.5% vs. 63.0%) and higher mAP@50:95 (36.7% vs. 32.5%), together with a slightly stronger precision-recall balance. This local result differs from the earlier Roboflow comparison from Sprint 2, which suggests that the relative ranking between YOLO11 and YOLO26 is sensitive to the exact training environment, augmentation pipeline, and evaluation protocol.

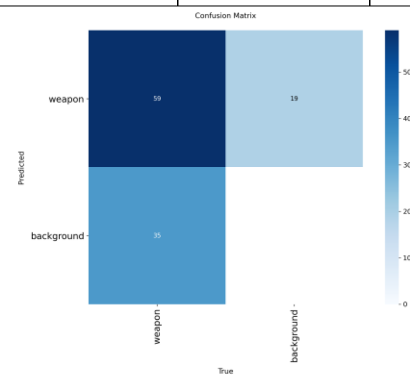
## 4. Error Analysis with Validation Confusion Matrices

The validation confusion matrices reinforce the metric comparison. YOLO11n produced 57 true positives, 21 false positives, and 37 false negatives. YOLO26n produced 59 true positives, 19 false positives, and 35 false negatives. In other words, YOLO26n slightly improved all three error indicators: it detected more true weapon instances while also producing fewer false alarms and fewer missed detections.

| Model | TP | FP | FN | Precision (derived) | Recall (derived) | F1 (derived) |
|---|---|---|---|---|---|---|
| YOLO11n | 57 | 21 | 37 | 73.1% | 60.6% | 66.3% |
| YOLO26n | 59 | 19 | 35 | 75.6% | 62.8% | 68.6% |




*YOLO11n validation confusion matrix*



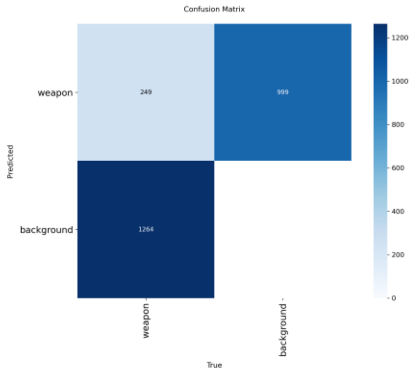

*YOLO26n validation confusion matrix*

## 5. Test-set Evaluation Artifacts

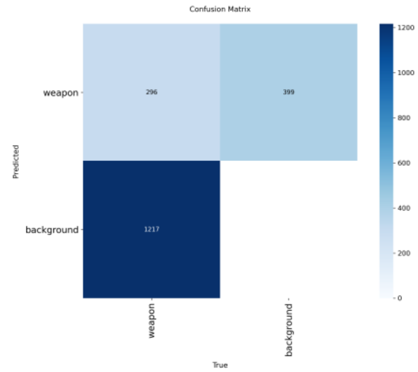
Additional test-evaluation artifacts were generated for both models on the official test partition. The available exported files include confusion matrices and qualitative prediction batches. Although the scalar test metrics were not exported together with the provided artifacts, the confusion matrices already show a clear difference in behavior between the two models.

| Model | TP | FP | FN | Precision (derived) | Recall (derived) | F1 (derived) |
|---|---|---|---|---|---|---|
| YOLO11n | 249 | 999 | 1264 | 20.0% | 16.5% | 18.0% |
| YOLO26n | 296 | 399 | 1217 | 42.6% | 19.6% | 26.8% |

On the provided test-evaluation artifacts, YOLO26n remained the stronger model. It produced more true positives (296 vs. 249), much fewer false positives (399 vs. 999), and fewer false negatives (1217 vs. 1264). Even though these figures were derived from confusion matrices rather than from an exported metric file, they strongly suggest that YOLO26n generalizes better to the official test split under the current setup.




*YOLO11n test confusion matrix*




*YOLO26n test confusion matrix*

## 6. Preliminary Conclusion

Sprint 3 successfully moved the project from data preparation to actual baseline comparison. The full local runs showed that YOLO26n is the stronger baseline under the current protocol, outperforming YOLO11n both in validation metrics and in confusion-matrix error profiles. The test-evaluation artifacts also favor YOLO26n, especially because it greatly reduces false positives while slightly increasing true positives.

The next step is to consolidate scalar test metrics, expand qualitative error analysis with representative prediction examples, and decide whether Sprint 4 should continue with YOLO26n as the main baseline or introduce a paper-faithful comparison model for broader methodological coverage.